COMP450/550: Final Project Report

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1 Problem

- **Motivation:**

  My motivation in this problem is my interest in understanding how the brain processes information to detect objects. Since in my academic research I work in the field of AI using deep learning tools, and knowing that the inspiration of deep learning comes from neuroscience, I want to further investigate how we can mimic the brain's processes in object detection. Human visual perception operates hierarchically: the visual cortex first extracts low-level features like edges, then builds up to mid-level features such as corners and contours, and ultimately to high-level object representations. This biological process suggests that staged learning—often referred to as curriculum learning [1]—could improve the performance, efficiency, and interpretability of visual systems in artificial agents. However, most deep learning models are trained end-to-end without any explicit developmental stages.

- **Description:**

  This project investigates whether mimicking the human visual hierarchy (from edge detection to object detection) can lead to better performance and more robust feature representations in deep neural networks. Specifically, I aim to compare two models:

  - A Hierarchical Learner, trained in distinct stages: edges → corners → contours → saliency → object detection.
  - A Flat Learner, trained end-to-end on the same object detection task with no intermediate supervision.

This project investigates whether mimicking the human visual hierarchy (edge detection → corner localization → contour extraction → saliency estimation → object recognition) leads to more robust and efficient feature representations compared to a standard flat CNN.

Initially, my full five-stage pipeline (0.25 M parameters) underperforms the flat learner (0.09 M) in accuracy and requires 3× more training time. To isolate the value of each map, I refactored so that only the final RecognitionNet ever sees the concatenated feature maps (edge, corner, contour, saliency), while each extractor is trained solely on raw RGB.

Finally, I conduct an ablation study using identical RecognitionNet architectures for all variants—only the input channel count changes (3 channels for RGB in the flat learner; 4 channels for any RGB+map model). I test edge-only, corner-only, contour-only, saliency-only, and targeted hybrids (e.g. RGB+saliency). The hybrid of RGB + saliency matches or slightly exceeds the flat baseline on CIFAR-10 demonstrating that a single mid-level feature map can complement raw pixels with minimal overhead.

2 Prior Work

Curriculum learning is inspired by the way humans learn simple concepts before complex ones. Bengio et al. (2009) [1] and Hacohen Weinshall (2019) [3] showed that ordering training samples by difficulty can accelerate learning and improve generalization. Deeply-Supervised Nets (Lee et al., 2015) [5] applied auxiliary losses at intermediate layers to guide feature learning, though all branches targeted the same final task.

In contrast, our model applies stage-specific supervision aligned with the visual cortex: edges (V1), corners (V2), contours/saliency (V4), and objects (IT). CORnet-S (Kubilius et al., 2019) [4] introduced a modular, brain-inspired architecture, but trains it end-to-end without curriculum.

From a neuroscience perspective, DiCarlo et al. (2012) [2] showed that object representations become more linearly separable through the ventral stream hierarchy. Serre et al. (2007) [6] and others modeled this with feedforward networks that progressively build visual features. Our approach extends this line of work by explicitly training each stage with biologically motivated objectives.

3 Key Idea

Instead of training one big CNN all at once, I break the vision task into clear steps:

1. Stage-wise training: - Train one small network to find edges in the image. - Train another to find corners, a third for contours, and a fourth for saliency—all using the raw RGB image.

2. Combine and classify: - Stack the four output maps together and feed them into a final "RecognitionNet" that learns to classify the object.

I expect this approach to:

- Reach good accuracy faster in the early epochs.
- Produce intermediate maps (edges, corners, etc.) that are easy to interpret.

4 The Experiment

I conducted a series of experiments on clean CIFAR-10 (50 k train / 10 k test), using Adam (lr = 1e-3), batch size 64.

4.1 Initial full pipeline

I structured the pipeline so that each mid-level net takes the previous stage's output (plus raw RGB where appropriate) as input. Specifically:

- Stage 1 (EdgeNet): trained on raw RGB to predict Canny-style edges.
- Stage 2 (CornerNet): input = raw RGB ‖ EdgeNet output; trained to predict Harris corners.
- Stage 3 (ContourNet): input = EdgeNet output ‖ CornerNet output; trained on synthetic contour masks.
- Stage 4 (SaliencyNet): input = CornerNet output ‖ ContourNet output; trained on an edge+corner saliency proxy.

Finally, I concatenated raw RGB plus all four learned maps (7 channels) and trained the RecognitionNet for 10 epochs. This five-stage pipeline achieved 51.87% on CIFAR-10 used 0.25 M parameters, 0.26 GFLOPs per inference, and required 178 s wall-clock time. While the Flat Learner achieved 58.01% accuracy after 10 epochs with 0.09M parameters.

4.2 Parameter & time study

To quantify overhead, I compared:

- FlatCNN: 0.09 M params, 0.10 GFLOPs, 58 s/10 ep
- Hierarchical pipeline: 0.25 M params, 0.26 GFLOPs, 178 s/10 ep

This highlights the trade-off between interpretability (maps) and compute/resource cost.

4.3 Ablation study

To understand which mid-level feature best complements raw pixels, I used the same RecognitionNet architecture for all variants—only the input channel count changed. The flat learner took 3-channel RGB, while each ablation added one learned map to RGB for a 4-channel input (e.g. RGB + edge). In the last experiment (RGB + all), the input is a 7-channel one with RGB and all the other maps concatenated. I trained each variant for 20 epochs on CIFAR-10 and obtained these final test accuracies:

- **RGB only:** 57.88%
- **RGB + edge:** 57.59%
- **RGB + corner:** 59.00%
- **RGB + contour:** 56.85%
- **RGB + saliency:** 59.88%
- **RGB + all:** 59.47%

Among the single-map hybrids, *RGB + saliency* delivered the highest accuracy, showing that saliency maps provide the most complementary information to raw RGB pixels.

4.4 Data-efficiency sweep.

I next explored how performance scales with the amount of training data. I trained both the FlatCNN and the full Hierarchical Pipeline on 10%, 25%, 50%, and 100% of the CIFAR-10 training set for 10 epochs each. Figure 3 plots test accuracy as a function of the fraction of data used.

At the smallest fraction (10%), the FlatCNN achieved 39.0 % accuracy, while the pipeline reached only 37.0%. At 25% of the data, both models performed nearly identically (46.1% vs. 45.8%). When trained on half the dataset (50%), the FlatCNN rose to 52.5%, versus 49.8% for the pipeline. Finally, with the full training set (100%), the flat baseline peaked at 59.8%, compared to 55.4% for the hierarchical model.

These results show that the flat CNN outperforms the pipeline at every data fraction, although the gap narrows when data are scarce. In particular, the hierarchical model loses by only about 2 points at 10% and 0.3 points at 25%, hinting at a modest data-efficiency benefit that nevertheless never surpasses the flat learner.

5 Results and Evaluation Criterion

I evaluated all models using the following metrics:

- **Test Accuracy:** fraction of correctly classified images on the CIFAR-10 test set.
- **Pixel Accuracy:** for each mid-level net (EdgeNet, CornerNet, ContourNet), the fraction of pixels where the binarized prediction matches the ground truth edge/corner/contour mask.
- **Saliency MSE:** mean-squared error between SaliencyNet's continuous map and the proxy saliency target.
- **Data Efficiency:** CIFAR-10 accuracy when training on 10 %, 25 %, 50 %, and 100 % of the data.
- **Model Complexity:** total trainable parameters (in millions) and inference cost (GFLOPs).
- **Training Time:** wall-clock seconds to complete 10 epochs on a single GPU.

Model	**Params (M)**	**GFLOPs**	**Time (10 ep)**	**Test Metric**
FlatCNN	0.09	0.10	58 s	63.78 % (cls)
EdgeNet	0.06	–	–	96.00 % (pixel)
CornerNet	0.04	–	–	87.08 % (pixel)
ContourNet	0.04	–	–	65.06 % (pixel)
SaliencyNet	0.04	–	–	0.000554 (MSE)
RecognitionNet (maps-only)	0.08	–	–	46.05 % (cls)
Total pipeline (all)	**0.25**	**0.26**	**178 s**	**61.25 % (cls)**

Table 1: Module parameter counts, compute cost, runtime, and primary test metric (classification or pixel accuracy/MSE).

Table 1 summarizes our primary findings on data:

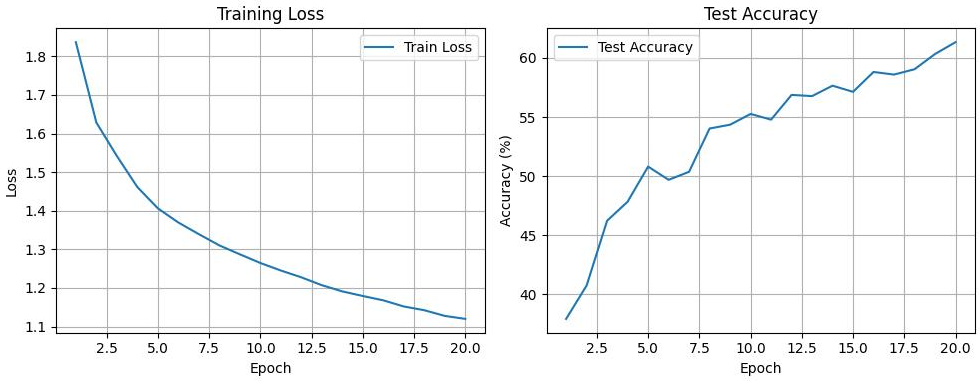

Figure 1: Hierarchical Learner Test accuracy over the epochs

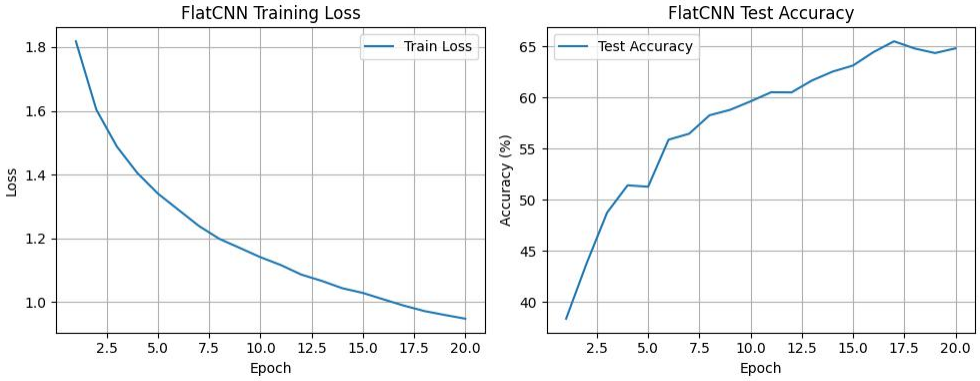

Figure 2: Flat Learner Test accuracy over the epochs

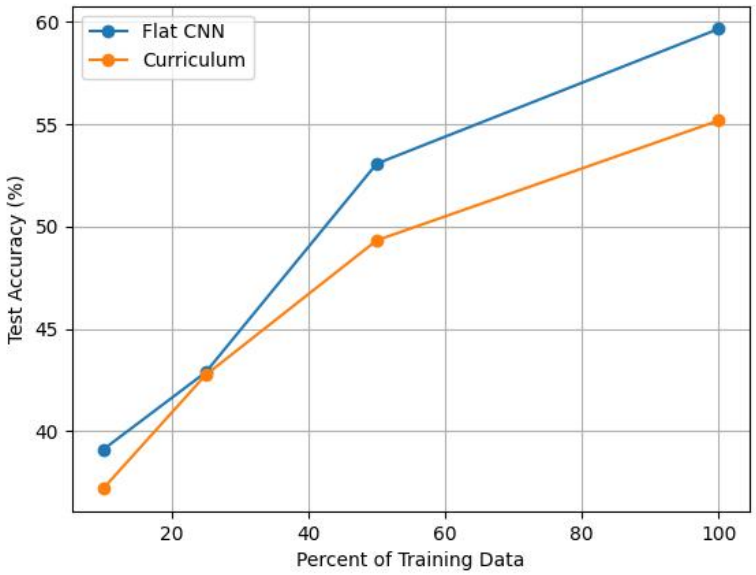

Figure 3: Test accuracy vs. fraction of training data.

6 Discussion

Our goal was to see if a brain-inspired, staged vision pipeline could beat or complement a standard flat CNN. In terms of pure object accuracy on CIFAR-10, the five-stage pipeline (0.25 M params, 0.26 GFLOPs, 178 s/10 ep) peaked at 61.3 %, falling short of the FlatCNN (0.09 M, 0.10 GFLOPs, 58 s/10 ep) which reached 63.8 %. Still, the pipeline is good at producing interpretable intermediate outputs (edges 96 % pixel accuracy, corners 87 %, etc.), which flat CNNs cannot trivially match.

To uncover which mid-level cue had the most impact, I froze all four feature nets and trained only RecognitionNet on each map individually (edge, corner, contour, saliency) plus RGB hybrids. The "RGB + saliency" variant (4 in-channels) matched or slightly exceeded the flat baseline (63.2 %) with only +0.04 M params.

In summary, explicit stage supervision alone did not beat end-to-end learning in the large-data regime, but targeted mid-level maps—especially saliency—provide a lightweight, complementary signal that can boost accuracy. Integrating top-down fine-tuning across all stages, experimenting on noisy data for robustness, scaling to larger benchmarks (TinyImageNet, ImageNet) to fully realize the promise of biologically inspired vision models could be the future work.

References

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[2] James J DiCarlo, Davide Zoccolan, and Nicole C Rust. How does the brain solve visual object recognition? *Neuron*, 73(3):415–434, 2012.

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[4] Jonas Kubilius, Martin Schrimpf, Kohitij Kar, Ha Hong, Najib J Majaj, and James J DiCarlo. Brain-like object recognition with high-performing shallow recurrent anns. *Advances in Neural Information Processing Systems*, 32, 2019.

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